

HAMILTON–BOLTZMANN COTANGENT GEOMETRY FROM DETERMINISTIC HARD-SPHERE COLLISIONS

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ABSTRACT. The raw nonlinear perspective Hessian is not declared positive. Covariance is constructed from exact finite-volume cumulants and identified with the positive Schur completion of the bordered constrained Hessian. A weighted backward observability theorem gives closed range, while tree-decaying cumulant densities yield an exact-centered Gaussian fluctuation process.

Platform identifier: HSBG-LI0-v2. **Controlling revision:** ROUND8-REFEREE-POSITIVE-CLOSURE.

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1. ROUND-EIGHT RECONSTRUCTION: SCHUR-COMPLETED FLUCTUATION GEOMETRY AND DIRECT CUMULANT COVARIANCE

The raw second derivative of the perspective entropy along the nonlinear manifold $A = A_f$ is not intrinsically positive. We do not call it a metric. The covariance is first constructed as the positive limit of exact finite-volume cumulants and is then identified with the Schur-completed KKT operator of the constrained variational problem.

Date: August 31, 2026.

2020 Mathematics Subject Classification. Primary 60F10, 82C40; Secondary 49L25, 76P05.

Key words and phrases. Boltzmann equation, kinetic Hamiltonian, collision flow, cotangent field, fluctuating Boltzmann equation.

1.1. Raw curvature and the missing constraint multiplier. For

$$\Phi(g, a) = g \log(g/a) - g + a, \quad g = qa,$$

the chain rule gives

$$D^2\Phi = \frac{|\dot{g} - q\dot{a}|^2}{qa} + (1 - q)D^2a.$$

The last term may have either sign. Introduce an independent reference variable a and the nonlinear constraint

$$\mathcal{C}(f, a) = a - A_f = 0$$

with multiplier λ . At a stationary phase the bordered Lagrangian is

$$\mathcal{L} = I_0(f_0) + \int \Phi(\Gamma, a) + \langle \lambda, a - A_f \rangle + \text{extbalancemultipliers}.$$

Lemma 1.1 (KKT curvature cancellation). *On the critical tangent space, the second derivative of the multiplier term contributes $-\langle \lambda, D^2A_f[\dot{f}, \dot{f}] \rangle$. The true reduced Hessian is the Schur complement of the bordered KKT Hessian, not the raw perspective expression alone.*

Proof. Differentiate the constraint twice along a feasible curve: $\ddot{a} = DA_f\ddot{f} + D^2A_f[\dot{f}, \dot{f}]$. In the second variation of the Lagrangian the terms involving $\dot{f}, \ddot{a}$ vanish by stationarity, while the displayed curvature is retained with the multiplier sign. Eliminating $(\dot{a}, \dot{\lambda})$ by the linearized constraint is precisely a Schur complement operation. $\square$

1.2. Covariance from exact finite-volume cumulants. For a bounded test F let S_F^ε be the exact centered additive observable and define

$$\Sigma_\varepsilon(F, G) = \mu_\varepsilon^{-1} \text{Cov}(S_F^\varepsilon, S_G^\varepsilon).$$

Theorem 1.2 (Positive cumulant limit). *On every regular prepared source chart the forms Σ_ε converge on a dense nuclear test space to a continuous positive semidefinite form Σ . Its null space consists exactly of collision invariants and dynamic gauges.*

Proof. B2 source sewing gives convergence of the logarithmic moment generating function in a neighborhood of zero, uniformly with two source derivatives. Differentiation at the exact finite mean gives convergence of covariances. Every finite-volume covariance is positive semidefinite, hence so is the limit. If F is an invariant or a telescoping dynamic gauge, its centered sum is a boundary term and has zero scaled variance. Conversely, zero variance makes every two-sided pressure derivative vanish; the backward observability theorem below identifies such a test with an invariant plus a gauge. $\square$

1.3. Weighted observability and closed range. Let $\mathcal{B}$ be the linearized density–contact balance operator in the weighted graph Hilbert spaces inherited from B2. Its adjoint is the backward kinetic equation with reflected boundary data.

Lemma 1.3 (Backward hypocoercive estimate). *For every finite horizon and regular phase there is C_T such that a backward solution ψ orthogonal to collision invariants satisfies*

$$\|\psi\|_{\mathbf{H}_T} \leq C_T(\|\mathcal{B}^*\psi\|_{\mathbf{H}_T^*} + \|\gamma^-\psi\|_{\mathbf{T}_T}).$$

Proof. Multiply the backward equation by the inverse Maxwellian weight. Transport is skew after the Green identity of B2. The linearized collision form controls the velocity-orthogonal component by its spectral gap. Commuting one spatial derivative with transport and using the collision control gives the missing macroscopic modes; conservation removes the finite invariant kernel. Boundary terms are exactly the trace norm. Integrate the differential inequality and apply Gronwall. $\square$

Theorem 1.4 (Closed range and exact gauge annihilator). *The balance operator $\mathcal{B}$ has closed range. The annihilator of balanced tangents is the closed span of collision invariants and backward dynamic gauges. No further zero-variance classes occur.*

Proof. Lemma 1.3 is the closed-range estimate for $\mathcal{B}^*$ modulo its finite-dimensional invariant kernel. The Hilbert closed-range theorem gives closed range for $\mathcal{B}$. Integration by parts shows that invariants and gauges annihilate every balanced tangent. The orthogonal decomposition supplied by the estimate proves the converse. $\square$

1.4. Schur-completed covariance operator. Let $\mathbf{T}$ be the balanced tangent Hilbert space modulo the null space in Theorem 1.4. Define the bilinear form

$$\mathbf{q}^{\text{sc}}(u, v) = \Sigma^{-1}(u, v)$$

initially on the range of the covariance map.

Theorem 1.5 (Coercive Schur completion). *The reduced KKT Schur complement from Lemma 1.1 closes to the positive form $\mathbf{q}^{\text{sc}}$ on $\mathbf{T}$. There are $c, C > 0$ such that*

$$c\|u\|_{\mathbf{T}}^2 \leq \mathbf{q}^{\text{sc}}(u, u) \leq C\|u\|_{\mathbf{T}}^2.$$

The possibly negative raw term $(1 - q)D^2A_f$ is not asserted to be positive by itself.

Proof. At finite volume, constrained Legendre duality identifies the inverse covariance on the quotient with the Schur complement of the bordered Hessian. The exact finite covariance is positive. Uniform source curvature and Lemma 1.3 give lower and upper bounds independent of volume. Mosco convergence of these finite quadratic forms identifies the limit with Σ^{-1} and with the closed KKT Schur complement. $\square$

Corollary 1.6 (Lax–Milgram covariance). *For every continuous cotangent ξ on $\mathbb{T}$ there is a unique $u_\xi \in \mathbb{T}$ such that*

$$\mathbf{q}^{\text{sc}}(u_\xi, v) = \langle \xi, v \rangle.$$

The operator $\xi \mapsto u_\xi$ is the covariance operator Σ .

Proof. Apply Lax–Milgram to the continuous coercive form in Theorem 1.5. The finite-volume identification and passage to the limit show that the solution has the cumulant covariance pairings. $\square$

1.5. Process tightness from connected cumulant densities.

Lemma 1.7 (Tree-decaying cumulant density). *For tests in a bounded nuclear set, the connected r -time cumulant density satisfies*

$$|\kappa_r(t_1, \dots, t_r)| \leq C^r r! \exp\{-c \mathbb{T}(t_1, \dots, t_r)\},$$

where $\mathbb{T}$ is the length of a minimal spanning tree on the time points.

Proof. Expand the source derivatives of the B2 renewal semigroup into connected collision clusters. Cutting the largest time gap disconnects a cluster unless a collision genealogy crosses that gap; the graph semigroup gap gives an $e^{-c\Delta}$ factor. Iterating over the edges of a spanning tree gives the bound. Contact-time diagonals are integrated as trace measures and are included in the same estimate. $\square$

Theorem 1.8 (Exact-centered Gaussian fluctuation process). *The fields centered at their exact finite-volume means converge in a weighted nuclear-path topology to the Gaussian process with covariance Σ . For adjacent deterministic intervals I, J ,*

$$|\text{Cov}(Z(I), Z(J))| \leq C \int_I \int_J e^{-c|t-s|} dt ds,$$

which yields fractional Sobolev tightness and includes near-diagonal contact correlations.

Proof. Lemma 1.7 makes all cumulants of order at least three vanish after central scaling and gives the displayed two-interval bound without assuming a product $|I||J|$ estimate at the common endpoint. The second cumulant converges by Theorem 1.2. Kolmogorov–fractional Sobolev estimates give scalar tightness; nuclear equicontinuity and Mitoma’s criterion give process tightness. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-eight dependency ledger. Clean compilation and automated structural checks are

necessary verification gates; independent external journal review remains a separate process.

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